

● MEDIUM

● **ADVANCED MATH**

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# Advanced Math

# 13

10 questions · ~15 min

Q1

ADVANCED MATH

$$14j + 5k = m$$

The given equation relates the numbers  $j$ ,  $k$ , and  $m$ . Which equation correctly expresses  $k$  in terms of  $j$  and  $m$ ?

- A.  $k = \frac{m - 14j}{5}$
- B.  $k = \frac{1}{5}m - 14j$
- C.  $k = \frac{14j - m}{5}$
- D.  $k = 5m - 14j$

A rubber ball bounces upward one-half the height that it falls each time it hits the ground. If the ball was originally dropped from a distance of 20.0 feet above the ground, what was its maximum height above the ground, in feet, between the third and fourth time it hit the ground?

STUDENT-PRODUCED RESPONSE

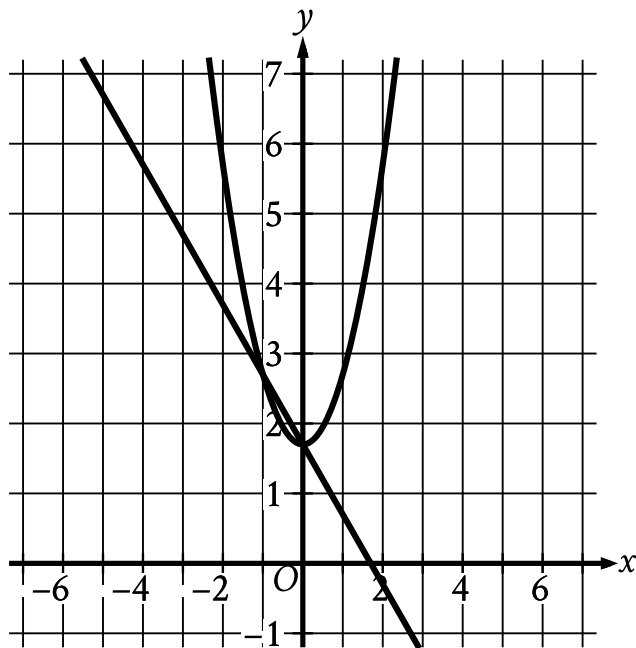
|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
| 1                    | 1                    | 1                    | 1                    |
| 2                    | 2                    | 2                    | 2                    |
| 3                    | 3                    | 3                    | 3                    |
| 4                    | 4                    | 4                    | 4                    |
| 5                    | 5                    | 5                    | 5                    |
| 6                    | 6                    | 6                    | 6                    |
| 7                    | 7                    | 7                    | 7                    |
| 8                    | 8                    | 8                    | 8                    |
| 9                    | 9                    | 9                    | 9                    |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = x^2 + 1.7$$

$$y = 1.7 - x$$

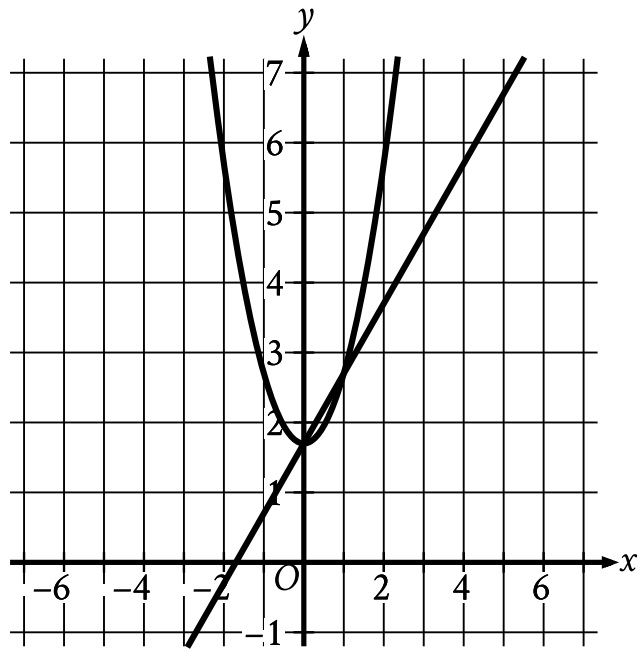
Which graph represents the given system of equations?



A.

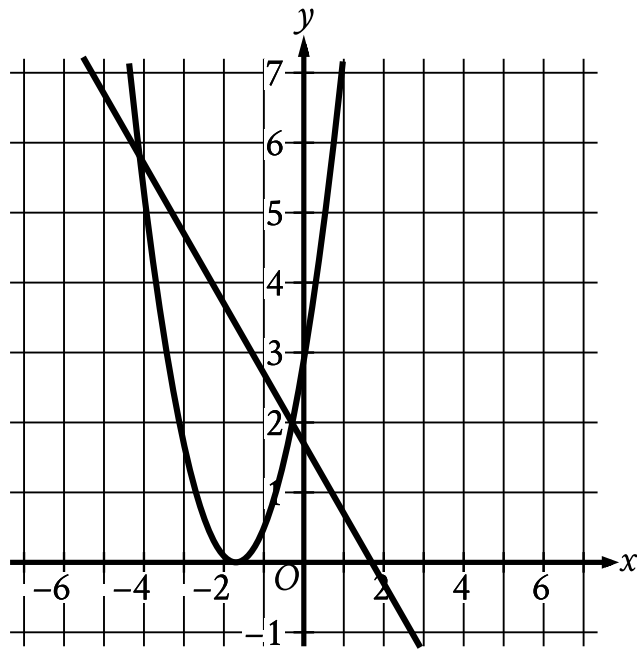
- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

B.



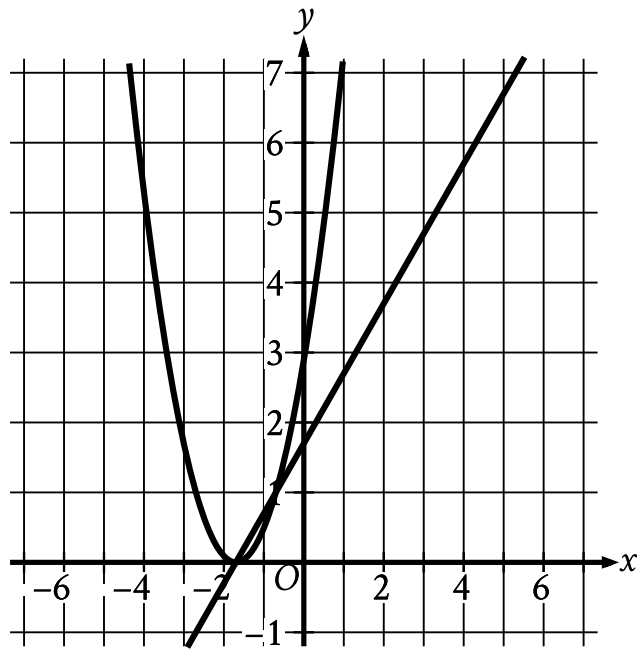
- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 1.7)
    - (1 comma 2.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

C.



- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (negative 1.7 comma 0).
  - The parabola passes through the following points:
    - (negative 2.7 comma 1)
    - (negative 1.7 comma 0)
    - (negative 0.7 comma 1)

D.



- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 1.7)
    - (1 comma 2.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (negative 1.7 comma 0).
  - The parabola passes through the following points:
    - (negative 2.7 comma 1)
    - (negative 1.7 comma 0)
    - (negative 0.7 comma 1)



$$m(t) = -0.0274\frac{t^2}{7} + 7.3873\frac{t}{7} + 75.032$$

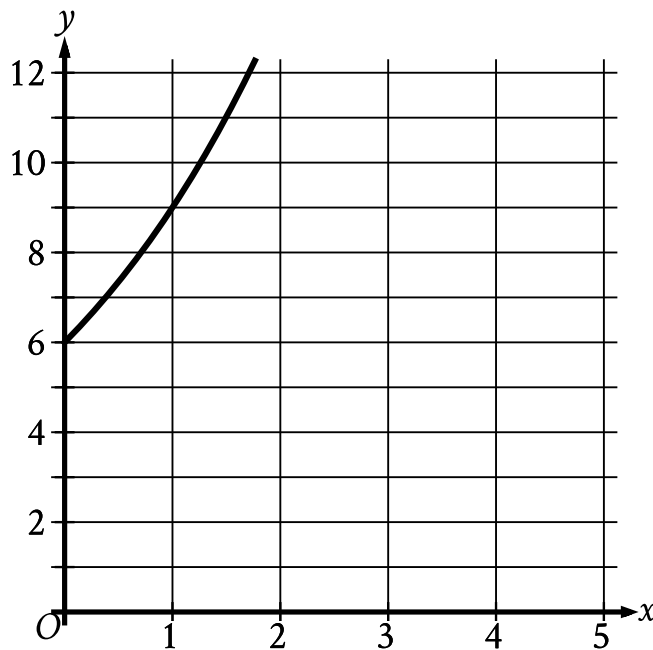
The function  $m$  gives the predicted body mass  $m(t)$ , in kilograms (kg), of a certain animal  $t$  days after it was born in a wildlife reserve, where  $t \leq 390$ . Which of the following is the best interpretation of the statement “ $m(330)$  is approximately equal to 362” in this context?

- A. The predicted body mass of the animal was approximately 330 kg 362 days after it was born.
- B. The predicted body mass of the animal was approximately 362 kg 330 days after it was born.
- C. The predicted body mass of the animal was approximately 362 kg  $\frac{330}{7}$  days after it was born.
- D. The predicted body mass of the animal was approximately  $\frac{330}{7}$  kg 362 days after it was born.

$$x^2 - x - 1 = 0$$

What values satisfy the equation above?

- A.  $x = 1$  and  $x = 2$
- B.  $x = -\frac{1}{2}$  and  $x = \frac{3}{2}$
- C.  $x = \frac{1+\sqrt{5}}{2}$  and  $x = \frac{1-\sqrt{5}}{2}$
- D.  $x = \frac{-1+\sqrt{5}}{2}$  and  $x = \frac{-1-\sqrt{5}}{2}$



- Moving from left to right:
  - The curve passes through quadrant 1.
  - The curve trends up sharply.
- The curve passes through the following points:
  - (0 comma 6)
  - (1 comma 9.0)

The graph gives the estimated population  $y$ , in thousands, of a town  $x$  years since 2003, where  $0 \leq x \leq 5$ . Which of the following best describes the increase in the estimated population from  $x = 0$  to  $x = 1$ ?

- A. The estimated population at  $x = 1$  is 0.5 times the estimated population at  $x = 0$ .
- B. The estimated population at  $x = 1$  is 1.5 times the estimated population at  $x = 0$ .
- C. The estimated population at  $x = 1$  is 2.5 times the estimated population at  $x = 0$ .

- D. The estimated population at  $x = 1$  is 3.5 times the estimated population at  $x = 0$ .

Q7

GRID-IN ADVANCED MATH

$$x + y = 17$$

$$xy = 72$$

If one solution to the system of equations above is  $(x, y)$ , what is one possible value of  $x$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A sample of a certain isotope takes 29 years to decay to half its original mass. The function  $s(t) = 184(0.5)^{\frac{t}{29}}$  gives the approximate mass of this isotope, in grams, that remains  $t$  years after a 184-gram sample starts to decay. Which statement is the best interpretation of  $s(87) = 23$  in this context?

- A. Approximately 23 grams of the sample remains 87 years after the sample starts to decay.
- B. The mass of the sample has decreased by approximately 23 grams 87 years after the sample starts to decay.
- C. The mass of the sample has decreased by approximately 87 grams 23 years after the sample starts to decay.
- D. Approximately 87 grams of the sample remains 23 years after the sample starts to decay.

If  $\frac{42}{x} = 7x$ , what is the value of  $7x^2$ ?

- A. 6
- B. 7
- C. 42
- D. 294

The function  $f$  is defined by  $fx = 8x^3 + 4$ . What is the value of  $f2$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.